altairsim — CP/M 2.2b on an Altair 8" floppy

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To launch it, name the machine file from inside this folder:

```
$ altairsim cpm22-buffered.toml
```

On **Windows** the program is `altairsim.exe`:

```
> altairsim.exe cpm22-buffered.toml
```

The machine boots itself — you type no `BOOT` command — and comes up at the CP/M prompt. Type `DIR` to see the disk:

```
56K CP/M 2.2b v2.3
For Altair 8" Floppy

A>DIR
```

Mike Douglas's track-buffered **CP/M 2.2b v2.3**, on an 8" Pertec FD-400 behind an 88-DCDD, booted by the DBL PROM at `FF00` — `RUN FF00` is the machine file's whole startup, because on a real disk Altair that was EXAMINE `FF00` and RUN.

`^E` (STOP) takes the keyboard back to the monitor at any point; `RUN` resumes. `^C` belongs to CP/M (it is warm boot) and CP/M gets it.

Settings changed for one run

The machine file runs CP/M at full speed, on the terminal where you started the program. To run the machine with a different setting, give the setting on the command line with `--set`. The machine file does not change. The text after `--set` is what you type after `SET` at the `altairsim>` prompt.

This command runs the machine at 2 MHz, the speed of a real Altair:

```
$ altairsim cpm22-buffered.toml --set "cpu0 clock_hz=2000000"
```

This command puts the console in a terminal window of its own:

```
$ altairsim cpm22-buffered.toml --set "sio0:a connect=terminal" --set "TERMINAL bsdel=bs"
```

You can give the three settings in one command. The chapter *Machines* of the User Manual describes `--set`.

The files

File	What it is
`cpm22-buffered.toml`	The machine: `base = "default"` plus the floppy in drive 0. Read it — it explains the memory arithmetic and what `readonly` really does on this controller.
`cpm22-terminal.toml`	The same machine and the same disk, but the console is a **built-in VT100 window** the simulator draws itself instead of stdio. The monitor stays in the terminal you launched from; CP/M comes up in its own window. No telnet client, no external emulator. Try `emulation=adm3a` for a period CP/M terminal.
`cpm22b23-56k.dsk`	The bootable system disk, built for a 56K machine. Carries `DDT.COM`, `M80`/`L80`, `MBASIC` and the `vi` text editor and the host-bridge utilities (`R`, `W`, `HDIR`, `SIMCLK`), with 6K free. Shared by both machine files above.

There is no undo. Drive 0 is mounted read/write because that is what a real machine is, and CP/M writes to `A:` for anything you create. In a clone `git checkout` puts the image back; in the package you were handed, nothing does. Copy it first if you are about to test writes in anger.

Three drives are empty. `MOUNT dsk0:drive1 "my-scratch.dsk"` fills one, or `FORMAT` from inside CP/M will make you a disk.

`BOOT.ASM` and `BIOS.ASM` — this CP/M's own, and the authoritative source the 88-DCDD was built from — are in `disks/mits-88dcdd/cpm22/buffered/`. They are source rather than product, so they stay in the repository and are not in the package.